

BUILD - TEST - EXPLAIN

Mobile Robotics Build Guide

Calibrate two IR sensors and program a line follower

LEVEL	FORMAT	EVIDENCE
Beginner	2 guided labs	Code, wiring, checks and quiz

This workbook supports the complete REES52 Academy course. Watch each linked lesson in the Academy, build with power disconnected, then record your evidence here.

Learner name: _____ Date: _____

Before you build

Materials checklist

ITEM	QTY	CHECK
Arduino Uno compatible board	1	[]
Two-channel IR line sensor set	1	[]
L298N motor driver	1	[]
Geared DC motors	2	[]
Robot chassis and wheels	1	[]
Battery pack with switch	1	[]
Jumper wires	12	[]

Safety and setup

- Keep the robot raised for the first motor test.
- Switch battery power off before changing motor wiring.
- Do not power motors directly from Arduino I/O pins.

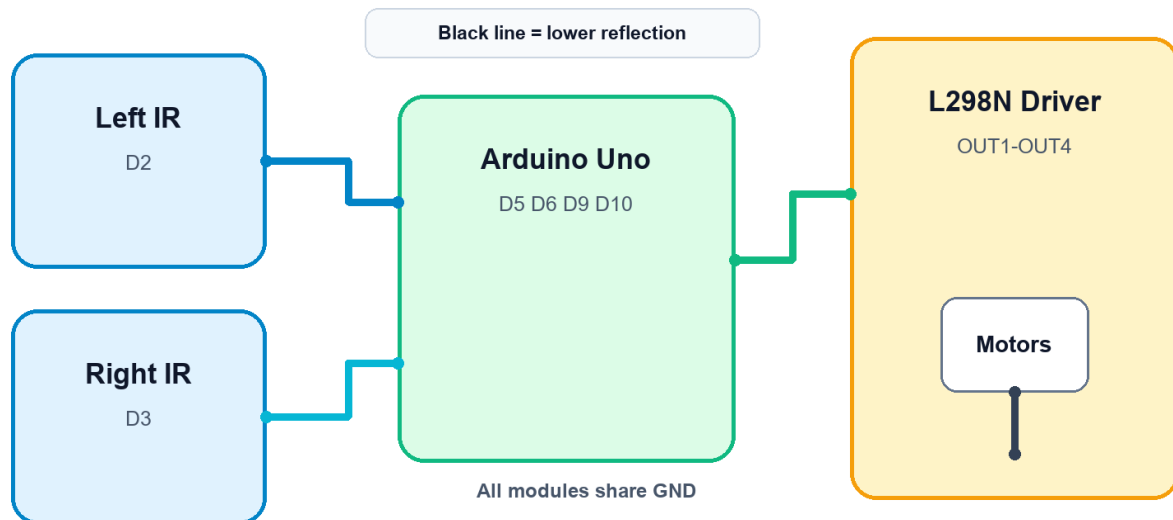
Evidence standard

A completed lab includes: a clear wiring photo, code that compiles, one observed result, one debugging note, and a short explanation in your own words.

Wiring reference

Two-Sensor Line Follower

Signal flow and motor-driver wiring



Verify every label against your board before applying power. Board variants can use different printed pin names.

Pre-power check

☐ Supply voltage checked ☐ Common ground connected ☐ No exposed wire strands ☐ Polarity confirmed

Lab 1: Calibrate the IR Sensors

Read both sensor outputs over light and dark surfaces, then record which digital state represents the line.

Build sequence

1. Mount both sensors at equal height above the test surface.
2. Connect left OUT to D2 and right OUT to D3.
3. Run the serial test and move each sensor across the line.
4. Adjust the onboard potentiometer until readings switch reliably.

Starter code

```
const int leftSensor = 2;
const int rightSensor = 3;

void setup() {
  pinMode(leftSensor, INPUT);
  pinMode(rightSensor, INPUT);
  Serial.begin(9600);
}

void loop() {
  Serial.print(digitalRead(leftSensor));
  Serial.print(",");
  Serial.println(digitalRead(rightSensor));
  delay(100);
}
```

Observation

What happened?

What did you change?

Explain why it worked:

Lab 2: Follow the Line

Map four sensor states to forward, left correction, right correction and stop.

Build sequence

1. Raise the chassis so the wheels can turn freely during first test.
2. Test each motor direction at low PWM before adding sensor logic.
3. Run the mapping and place the robot on a broad closed track.
4. Tune one speed value at a time and record the effect.

Starter code

```
void drive(int left, int right) {  
  analogWrite(5, constrain(left, 0, 255));  
  analogWrite(9, constrain(right, 0, 255));  
}  
  
void loop() {  
  int L = digitalRead(2);  
  int R = digitalRead(3);  
  if (!L && !R) drive(150, 150);  
  else if (L && !R) drive(70, 165);  
  else if (!L && R) drive(165, 70);  
  else drive(0, 0);  
}
```

Observation

What happened?

What did you change?

Explain why it worked:

Troubleshooting map

SYMPTOM	CHECK FIRST	NEXT TEST
No output	Power, board and selected port	Run the smallest known-good sketch
Unstable readings	Loose ground or long signal wires	Print raw values and average readings
Wrong direction	Motor or output mapping	Test one output channel at a time
Upload fails	USB cable and selected board	Close Serial Monitor and reconnect

Debugging record

Issue found: _____

Test performed: _____

Final fix: _____

Course check

Choose one answer for each question. Submit the online quiz in REES52 Academy for scoring and feedback.

1. What do the IR sensors measure?

A) Reflected infrared light B) Motor voltage only C) Wi-Fi strength D) Sound level

2. Why use a motor driver?

A) To handle motor current and direction B) To store PDFs C) To measure humidity D) To replace the battery

3. What should be tested first?

A) One motor direction at low speed B) Maximum speed on the floor C) Both batteries in reverse D) The certificate page

4. What does calibration improve?

A) Reliable line detection B) PDF page count C) Wi-Fi range D) Servo angle

5. How should speed be tuned?

A) One value at a time B) All values randomly C) Only by changing voltage D) Without observing the robot

Reflection

The concept I can now explain clearly is:

My next improvement will be:

Completion evidence

Use this final page before marking the course complete.

EVIDENCE	READY
All lesson videos watched	[]
Wiring built and photographed	[]
Starter code modified and saved	[]
Measured result recorded	[]
Online quiz passed	[]
Project explanation written in learner's own words	[]

Video source

The course lessons use tutorials published by REES52 at <https://rees52.com/pages/tutorials>. Links are also available inside each Academy lesson.

Course completion certificates show learning progress inside REES52 Academy. They are not a degree, licence, or government-accredited qualification.